

MALBORK, Poland

WMO: 121540

Lat: 54.027N

Lon: 19.134E

Elev: 16

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.2	8.5	-2.6	4.8	2.8	3.3	6.5	10.6	29.7	42.9	25.7	39.0	3.2	70	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	84.8	68.1	81.3	66.3	78.4	65.1	70.7	80.3	68.7	77.4	66.9	75.0	8.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.6	101.3	75.1	65.8	95.2	72.8	63.9	89.0	70.6	34.6	80.5	32.9	77.3	31.4	75.0	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	19.5	17.1	DB	-4.0	91.7	10.0	4.8	-11.2	95.2	-17.0	98.1	-22.7	100.8	-30.0	104.3	(4)
(5)			WB	-4.1	74.1	10.2	2.5	-11.4	75.9	-17.3	77.3	-23.1	78.7	-30.4	80.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	28.1	30.4	37.1	46.2	55.6	61.8	66.1	65.3	58.0	48.3	40.3	34.0	(6)	
	DBStd	14.92	10.60	9.52	8.20	6.72	6.92	5.83	5.16	4.87	5.18	6.34	5.83	8.50	(7)	
	HDD50	2703	680	547	403	150	21	0	0	0	4	110	291	497	(8)	
	HDD65	6521	1145	967	866	564	306	131	48	56	218	518	740	962	(9)	
	CDD50	1862	0	0	3	36	194	353	500	473	245	57	1	0	(10)	
	CDD65	202	0	0	0	0	14	33	83	64	8	0	0	0	(11)	
	CDH74	1861	0	0	0	12	151	353	699	566	80	0	0	0	(12)	
	CDH80	456	0	0	0	1	22	85	187	145	16	0	0	0	(13)	
(14) Wind	WSAvg	8.0	9.2	8.5	8.5	7.8	6.9	7.0	6.9	7.1	7.2	8.1	8.8	9.6	(14)	
(15) Precipitation	PrecAvg	25.1	1.5	1.3	1.3	1.3	2.3	2.5	3.1	3.0	2.6	2.5	2.0	1.7	(15)	
	PrecMax	31.3	3.0	2.4	2.6	3.1	3.6	4.8	6.2	5.9	6.9	5.1	4.1	3.0	(16)	
	PrecMin	20.6	0.4	0.6	0.4	0.1	0.9	0.3	0.6	0.7	0.8	0.5	0.6	0.7	(17)	
	PrecStd	2.8	0.7	0.4	0.6	0.8	0.8	1.3	1.3	1.3	1.5	1.3	0.9	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.8	50.1	62.4	75.0	82.7	87.9	89.8	89.4	82.1	68.2	54.7	51.6	(19)	
		MCWB	46.0	45.9	51.1	57.2	64.7	71.4	69.8	69.4	66.5	59.2	50.9	48.7	(20)	
	2%	DB	46.1	47.3	56.4	68.6	78.6	81.8	84.8	83.9	75.5	63.9	52.3	48.4	(21)	
		MCWB	43.2	43.5	46.9	55.0	64.9	67.1	68.4	67.6	62.5	57.1	49.5	45.9	(22)	
	5%	DB	43.3	44.3	52.2	63.4	73.8	77.8	81.0	80.1	71.5	60.8	50.2	46.3	(23)	
		MCWB	40.9	40.7	45.5	51.2	61.4	63.8	67.0	65.4	60.3	55.7	47.8	43.9	(24)	
	10%	DB	40.7	41.4	48.6	59.2	69.4	74.2	77.6	76.5	68.1	57.6	48.4	44.2	(25)	
		MCWB	38.8	38.6	43.4	49.2	58.8	62.1	65.8	64.1	58.9	53.4	46.3	42.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.8	47.0	52.7	60.8	67.7	73.5	73.4	72.3	67.4	60.9	52.1	49.3	(27)	
		MCDB	49.3	49.2	60.4	73.3	78.0	85.6	83.1	84.1	77.7	66.0	53.6	51.3	(28)	
	2%	WB	43.6	43.5	48.9	55.7	65.0	69.2	71.1	69.5	64.3	58.2	50.1	46.4	(29)	
		MCDB	45.7	46.7	53.6	65.4	76.2	78.7	80.1	78.8	72.9	62.4	51.9	48.2	(30)	
	5%	WB	41.0	40.9	46.1	52.7	62.6	66.1	69.2	67.5	61.8	55.9	48.1	44.0	(31)	
		MCDB	43.0	43.5	50.8	61.7	72.2	74.5	77.5	75.5	68.1	59.9	49.7	45.7	(32)	
	10%	WB	38.7	38.7	43.6	50.3	59.9	63.5	67.3	65.8	60.0	54.1	46.5	41.9	(33)	
		MCDB	40.6	41.2	48.2	57.3	67.2	70.7	75.1	73.4	66.5	57.4	48.2	43.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	10.3	14.3	19.6	19.6	19.5	18.1	18.8	16.6	12.9	8.2	7.6	(35)	
		MCDBR	7.9	11.9	20.6	28.0	25.4	25.9	24.3	24.8	23.2	17.2	10.1	8.9	(36)	
	5% WB	MCWBR	7.0	9.3	13.1	15.9	13.0	12.9	10.9	10.9	11.9	11.1	8.4	7.9	(37)	
		MCWBR	7.8	10.6	18.2	24.5	23.3	23.5	20.8	21.0	19.9	15.5	9.6	8.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.307	0.338	0.369	0.416	0.406	0.402	0.436	0.424	0.381	0.362	0.346	0.307	(40)	
		taud	2.243	2.255	2.256	2.179	2.266	2.306	2.234	2.271	2.370	2.410	2.318	2.270	(41)	
	Ebn at Noon		201	235	254	256	267	269	256	252	249	226	185	176	(42)	
		Edh at Noon	19	27	33	41	40	39	41	37	30	23	19	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		169	387	778	1324	1696	1772	1602	1387	991	516	198	121	(44)	
	RadStd		26	58	99	166	168	154	184	133	114	79	26	15	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (26 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-252	+126	N/A

Nomenclature: See separate page